

Clinical Investigation: Central Nervous System Tumor

# Dosimetric and Late Radiation Toxicity Comparison Between Iodine-125 Brachytherapy and Stereotactic Radiation Therapy for Juxtapapillary Choroidal Melanoma

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Received Nov 1, 2012, and in revised form Dec 22, 2012. Accepted for publication Feb 5, 2013

## Summary

Stereotactic radiation therapy provides a noninvasive treatment alternative for juxtapapillary choroidal melanoma. We have observed significant late ocular toxicity rates with this technique. We study here whether these high toxicities are related to the tumor juxtapapillary location or the stereotactic radiation therapy technique, through comparing the toxicity rates and their potential correlation to the intraocular radiation dose distributions, between 2 highly comparable cohorts of patients, one treated with the conventional brachytherapy and the other with stereotactic radiation therapy.

**Purpose:** To compare the dose distributions and late radiation toxicities for <sup>125</sup>I brachytherapy (IBT) and stereotactic radiation therapy (SRT) in the treatment of juxtapapillary choroidal melanoma.

**Methods:** Ninety-four consecutive patients with juxtapapillary melanoma were reviewed: 30 have been treated with IBT and 64 with SRT. Iodine-125 brachytherapy cases were modeled with plaque simulator software for dosimetric analysis. The SRT dosimetric data were obtained from the Radionics XKnife RT3 software. Mean doses at predetermined intraocular points were calculated. Kaplan-Meier estimates determined the actuarial rates of late toxicities, and the log-rank test compared the estimates.

**Results:** The median follow-up was 46 months in both cohorts. The 2 cohorts were balanced with respect to pretreatment clinical and tumor characteristics. Comparisons of radiation toxicity rates between the IBT and SRT cohorts yielded actuarial rates at 50 months for cataracts of 62% and 75% ( $P = .1$ ), for neovascular glaucoma 8% and 47% ( $P = .002$ ), for radiation retinopathy 59% and 89% ( $P = .0001$ ), and for radiation papillopathy 39% and 74% ( $P = .003$ ), respectively. Dosimetric comparisons between the IBT and SRT cohorts yielded mean doses of 12.8 and 14.1 Gy ( $P = .56$ ) for the lens center, 17.6 and 19.7 Gy ( $P = .44$ ) for the lens posterior pole, 13.9 and 10.8 Gy ( $P = .30$ ) for the ciliary body, 61.9 and 69.7 Gy ( $P = .03$ ) for optic disc center, and 48.9 and 60.1 Gy ( $P < .0001$ ) for retina at 5-mm distance from tumor margin, respectively.

**Conclusions:** Late radiation-induced toxicities were greater with SRT, which is secondary to the high-dose exposure inherent to the technique as compared with IBT. When technically feasible, IBT is preferred to treat juxtapapillary choroidal melanoma. © 2013 Elsevier Inc.

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Conflict of interest: none.

## Introduction

Treatment of choroidal melanoma adjacent to the optic nerve (juxtapapillary) using conventional round radioactive plaques can be technically challenging. The optic nerve is anatomically wider in its extraocular cross section as compared with the intraocular optic disc, which can be a physical obstacle to completely covering the tumor base bordering the optic disc. The optic nerve may cause the posterior plaque edge to slide along the nerve and tilt away, which compromises tumor coverage (1-3). Moreover, plaque insertion for tumors in this location carries a potential risk for mechanical injury of the optic nerve and its vasculature that may cause traumatic visual loss.

Various radiation techniques have been used specific to juxtapapillary choroidal melanoma and include the application of notched and slotted radioactive plaques, charged particle radiation therapy, single-fraction radiosurgery, and external beam stereotactic radiation therapy (SRT) (4-10). At our institution, SRT is the usual treatment for these tumors; however, some patients who declined, or were physically unsuitable for, SRT were treated with  $^{125}\text{I}$  brachytherapy (IBT), provided that plaque radiation therapy preplanning demonstrated the potential for adequate tumor coverage.

In a previous report we observed relatively high incidence rates of late radiation-induced toxicities with SRT for juxtapapillary tumors (4). This observation has prompted further investigation of whether SRT is in fact the appropriate therapy for such tumors. Therefore, in this report we focus on the comparison of the toxicity rates and the physical dose distribution profiles between 2 highly comparable cohorts of patients with juxtapapillary melanoma, one group treated with IBT and the other with SRT.

## Methods and Materials

We conducted a retrospective chart review for all consecutive patients with juxtapapillary choroidal melanoma treated with radiation therapy between October 1998 and January 2006 and subsequently followed in our institution. Approval of the research ethics board of the institution was obtained for chart review.

Tumors were defined as juxtapapillary if the posterior tumor margin was  $\leq 2.5$  mm beyond the optic disc. This is because the optic nerve is 1.5-2 mm larger in diameter on the episcleral side from the intraocular optic disc; and because the round episcleral plaques used are designed to be 3-4 mm larger in diameter than the largest tumor diameter, all plaques inserted to treat choroidal melanoma located at  $\leq 2.5$  mm from the optic disc should be virtually touching the episcleral portion of the optic nerve.

Patients were divided into 2 groups: patients treated with IBT and those treated with SRT. Patient data included demographics, medical and ocular history, and pretreatment findings based on clinical ophthalmologic examination, fundus photography, ultrasonography, and fluorescein angiography. The patients in both cohorts were followed at the same frequency: 3 months after radiation therapy for 2 visits, every 6 months for 5 years, and every 9-12 months thereafter.

Radiation-induced toxicities included radiation-induced cataract, neovascular glaucoma (NVG), retinopathy, and optic neuropathy. Radiation cataract was defined as the development of significant lens opacity in the irradiated eye as compared with the opposite eye (Fig. 1A). Neovascular glaucoma was diagnosed if there was neovascularization of the iris or angle associated with elevated

intraocular pressure (Fig. 1B). Radiation retinopathy was defined as the presence of clinical or angiographic evidence of spreading retinal ischemic vasculopathy more than 2 mm away from the treated tumor margin, with or without macular involvement, manifesting as retinal hemorrhage, exudates, infarcts, edema, or neovascularization. Presence of at least 2 of the previous findings was required to record the development of radiation retinopathy (Fig. 1C). Radiation papillopathy was defined as development of optic disc swelling with peripapillary hemorrhages and exudates, or disc pallor (Fig. 1D).

## Iodine-125 brachytherapy

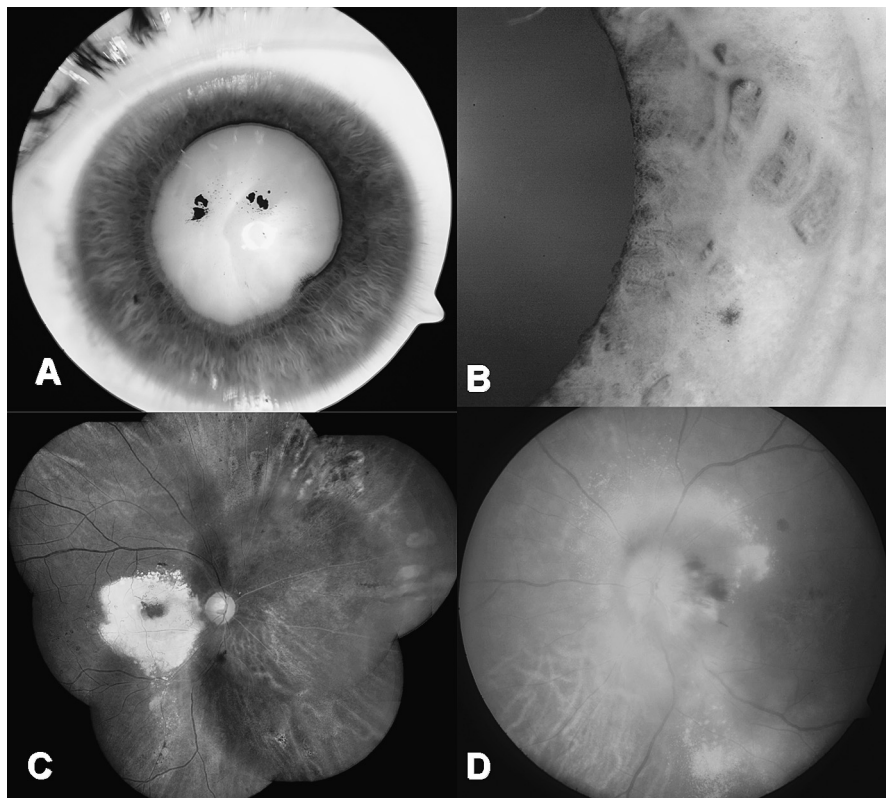
Brachytherapy was applied using the standard (Collaborative Ocular Melanoma Study, COMS) round, gold-shielded, episcleral plaques. The plaque diameter was at least 4 mm larger than the tumor's longest basal dimension. Guided by dosimetry plaque simulator software (Ekert & Ziegler BEBIG, Berlin, Germany), the  $^{125}\text{I}$  seeds were evenly loaded with single-activity seeds, and/or nonuniformly loaded with different activities seeds ranging from 0.5 to 7 mCi. Inserting higher-energy seeds at slots nearer to the posterior plaque edge would push the radiation penumbra more posteriorly and can include a thin posterior tumor margin adjacent to the optic disc within the radiation field. Because of plaque placement limitation, almost 1-mm margin coverage could be achieved at the nerve side, beyond plaque edge, with dose coverage on average approximately 15% lower than at tumor apex. According to COMS protocol, and the allocated operation times, all melanomas in this study were treated to a total dose of 8500 cGy delivered to the tumor apex at a mean dose rate of 50 cGy/h (range 49.5- 52.2 cGy/h) over 7 days. Plaques in this study were inserted under general anesthesia. Proper plaque placement was verified intraoperatively using an illuminated sclera indenter.

## Stereotactic radiation therapy

The prescribed total radiation dose was 70 Gy delivered in 5 fractions, with treatment every alternate day over 10 days of 6-MV photons. All patients were CT-simulated supine and immobilized using the Radionics Gill-Thomas-Cosman stereotactic relocatable head frame (Integra Radionics, Burlington, MA) adapted with an in-house adjustable eye-monitoring device. The radiation therapists monitor eye position via the eye-monitoring device video camera, and the in-house-developed software registers the eye position and detects deviations. If deviation occurs, treatment is stopped, the patient is allowed to rest, and treatment then restarts. Treatment planning was accomplished using the Radionics XKnife-4 and more recently XKnife RT3 software (Integra Radionics) (Fig. 2). Multiple converging noncoplanar arcs of radiation were generated using circular collimators ranging from 10 to 27.5 mm. Radiation therapy was delivered using Radionics circular collimators and the linear accelerator couch-mount assembly system mounted on a Varian 2100C (Varian, Palo Alto, CA) linear accelerator, and more recently the Elekta Synergy S unit (Elekta, Crawley, United Kingdom).

## Dosimetry

Each IBT-treated tumor was retrospectively replanned with plaque simulator software, to obtain the 3-dimensional dosimetric data for evaluation. Tumors treated with SRT were replanned with the



**Fig. 1.** Radiation toxicities. (A) Radiation cataract; (B) iris neovascularization in neovascular glaucoma; (C) radiation retinopathy; (D) radiation papillopathy.

Radionics XKnife RT 3 software. The maximum dose to pre-determined anatomic intraocular points that included the lens center, lens posterior pole, ciliary body at 2 points (center of ipsilateral ciliary body and center of contralateral ciliary body), optic disc center, and retina at 2 diametrically opposite points (each point 5 mm beyond the nearest tumor margin) were recorded (Fig. 2). The ciliary body dose for each patient represents the mean of the 2 measured ciliary body doses, and the mean retina dose at 5 mm in each patient was the mean of the 2 measured retina doses. The cumulative doses at the center of tumor base, tumor apex, and tumor center (midway between the former 2 points) were also recorded.

## Statistical analysis

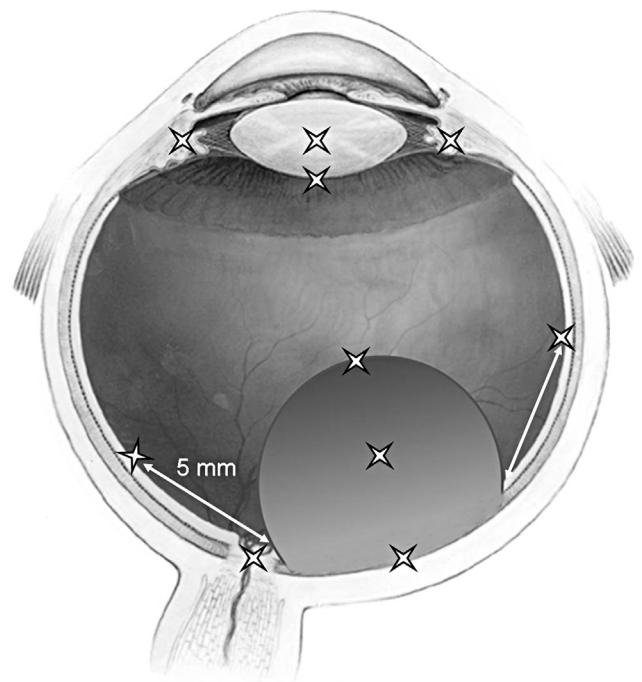
Patient overall survival was calculated according to Kaplan-Meier estimates. Rates of complications were calculated using the cumulative incidence approach. When the variables were continuous they were dichotomized at the median. Log-rank test was used to compare the statistical significance between the survival curves in both cohorts. Results were considered significant if the  $P$  value was  $\leq .05$ . Statistical analyses were performed using version 9.1 of the SAS system (SAS Institute, Cary, NC).

## Results

### Pretreatment patient data

The IBT cohort included 30 patients with the posterior tumor margin located 0-2.5 mm from the optic disc. The SRT cohort

included 64 patients with the posterior tumor margin located 0-2.1 mm from the optic disc. The 2 cohorts were found to be highly comparable, with no statistically significant differences with respect to baseline characteristics (Table 1).



**Fig. 2.** Illustration of the intraocular points at which the dosimetry values were obtained for each patient in both cohorts.

**Table 1** Comparison of the pretreatment patient and tumor characteristics between the  $^{125}\text{I}$  brachytherapy and stereotactic radiation therapy cohorts

| Characteristic                    | $^{125}\text{I}$ Iodine brachytherapy (n=30) | Stereotactic radiation therapy (n=64) | P   |
|-----------------------------------|----------------------------------------------|---------------------------------------|-----|
| Age (y)                           | 59 [58, 39-77]                               | 63 [62, 28-89]                        | .11 |
| Sex (male:female)                 | 15 (50):15 (50)                              | 36 (56):28 (44)                       | .57 |
| Diabetes                          | 3 (10)                                       | 3 (5)                                 | .33 |
| Hypertension                      | 7 (23)                                       | 18 (28)                               | .62 |
| Tumor height (mm)                 | 4.3 [4.5, 1.5-9.1]                           | 4.2 [4.6, 1.5-11]                     | .98 |
| Tumor base (mm)                   | 10.0 [11.0, 6-16]                            | 9.9 [10.2, 4.7-17]                    | .18 |
| Distance: tumor edge to disc (mm) | 0-2.5                                        | 0-2.1                                 |     |
| Relationship: tumor edge to disc  |                                              |                                       |     |
| Not touching the disc             | 15 (50)                                      | 22 (34)                               |     |
| >0° to <180°                      | 14 (47)                                      | 36 (57)                               |     |
| >180° to <360°                    | 1 (3)                                        | 4 (6)                                 |     |
| 360°                              | 0                                            | 2 (3)                                 |     |
| Follow-up (mo)                    | 46 [51, 6-105]                               | 46 [50, 6-118]                        | .46 |

Values are number (percentage) or median [mean, range], unless otherwise noted.

## Local control and toxicity

The actuarial rates were calculated at 50 months after treatment in both cohorts. Both cohorts were comparable with regard to tumor local control, with no statistically significant difference between rates of posttreatment local recurrence. Post-SRT rates of NVG, radiation retinopathy, and radiation papillopathy were significantly greater than in the IBT cohort; however, no difference in the rate of cataract was observed (Table 2).

## Dosimetric analysis

The mean cumulative physical dose delivered to each of the specified intraocular points between the 2 cohorts is illustrated in Table 3.

There was an exponential dose decline from the prescribed treatment dose to the doses delivered to the optic nerve, the retina at 5 mm from tumor, and the anterior segment structures. This dose fall-off was steeper in the IBT treated cohort than in those with SRT (Fig. 3).

## Discussion

Brachytherapy of juxtapapillary tumors carries risk of mechanical injury to the optic nerve (11). At our institution we opted for SRT as a noninvasive radiation therapy procedure to treat such tumors. Our approach was to deliver a total dose and fractionation scheme akin to those used in some proton beam radiation therapy

protocols (70 Gy in 5 fractions) (8, 12), but with the SRT-specific technology. We also treated on alternate days (delivering 5 fractions over 10 days), with the aim to maximize normal tissue repair between fractions (13). Theoretically SRT is designed to maximize dose to the tumor and minimize dose to surrounding tissues because of the steep dose gradient inherent to radiosurgery technology (14).

However, those juxtapapillary melanoma patients who declined SRT as an option, or were found to be physically and/or mentally incapable of undertaking the SRT procedures, were treated with IBT using the standard COMS plaques. This was with the caveat that IBT preplanning showed adequate coverage of the majority of the tumor, and we accepted relative underdosing at the tumor margin closest to the optic nerve head. Iodine-125 brachytherapy has the advantage of yielding a sharper dose gradient as compared with an external beam irradiation technique. Additionally, IBT has a comparative advantage owing to its low dose rate of continuous exposure that maximizes normal tissue repair (13). Therefore in this report we are able to perform a unique comparison of IBT using standard round plaque with SRT, specific to the juxtapapillary location.

The direct relationship between the radiation dose delivered to various intraocular structures and the incidence of late radiation toxicities of these structures has been previously analyzed in a report on IBT for large choroidal melanoma (15). Similarly, we have retrospectively replanned the patients to obtain dosimetric data, considering that there may be a discrepancy between actual plaque position in a patient and treatment planning. A plaque displacement of 0.5 mm may introduce up to 10% error in dose at

**Table 2** Comparison of the actuarial rates of tumor recurrence and radiation toxicities between patients treated with  $^{125}\text{I}$  brachytherapy and stereotactic radiation therapy at 4 years after radiation therapy

| Parameter                  | $^{125}\text{I}$ Iodine brachytherapy (n=30) | Stereotactic radiation therapy (n=64) | Hazard ratio | P     |
|----------------------------|----------------------------------------------|---------------------------------------|--------------|-------|
| Tumor recurrence (%)       | 11 (3-39)                                    | 7 (3-18)                              | 1.47         | .61   |
| Radiation cataract (%)     | 62 (42-83)                                   | 75 (59-88)                            | 0.63         | .1    |
| Neovascular glaucoma (%)   | 8 (2-29)                                     | 47 (35-62)                            | 0.20         | .002  |
| Radiation retinopathy (%)  | 60 (41-79)                                   | 89 (78-96)                            | 0.29         | .0001 |
| Radiation papillopathy (%) | 40 (24-63)                                   | 74 (61-86)                            | 0.37         | .003  |

Values in parentheses are 95% confidence intervals.



**Table 3** Mean physical radiation dose (Gy) delivered to various intraocular structures in the  $^{125}\text{I}$  brachytherapy and stereotactic radiation therapy cohorts

| Structure                     | $^{125}\text{I}$ Iodine brachytherapy (n=30) | Stereotactic radiation therapy (n=64) |
|-------------------------------|----------------------------------------------|---------------------------------------|
| Tumor base                    | 224.6 (138-428.4)                            | 72.4 (57.7-77.7)                      |
| Tumor center                  | 132.0 (112-179.6)                            | 73.9 (68.4-78.1)                      |
| Tumor apex                    | 86.1 (64.4-143.5)                            | 74.4 (63.5-78.5)                      |
| Optic disc center             | 61.9 (27.0-114.4)                            | 69.7 (10.8-74.4)                      |
| Retina (5 mm from tumor edge) | 48.9 (27.4-97.1)                             | 60.1 (25.5-73.2)                      |
| Lens posterior pole           | 17.6 (11.6-38.2)                             | 19.7 (1.3-73.8)                       |
| Lens center                   | 12.8 (7.8-29.4)                              | 14.1 (1.0-73.3)                       |
| Ciliary body                  | 13.9 (7.1-37.9)                              | 10.8 (0.7-57.2)                       |

Values are mean dose (range). The absolute point doses cannot be compared effectively between these 2 dose delivery systems (see [Discussion](#)).

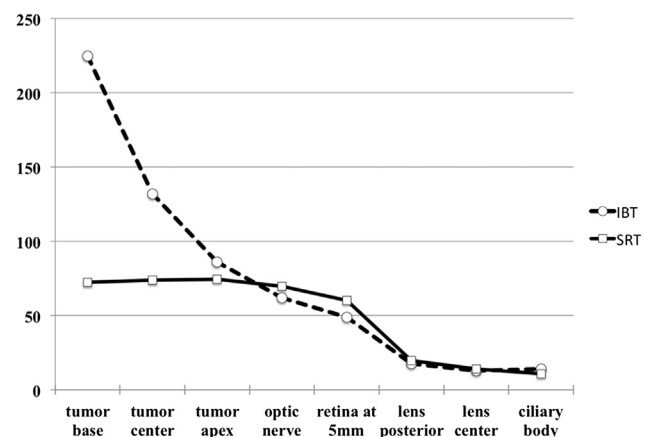
the edge of the plaque, which may also affect overall dosimetry results.

In this study we attempted to investigate the causes of higher toxicity rates associated with SRT in comparison with IBT, through analysis of intraocular radiation dose distributions. Nevertheless it is important to note that comparison of the radiation toxicity rates on the basis of their relation to the absolute physical doses between these 2 fundamentally radiobiologically different techniques is not valid. The biologic effect of the SRT and IBT dose distributions are to be considered independently, because existing models like the biologically equivalent dose may not apply to high-dose-per-fraction radiation as used in our SRT approach (14 Gy per fraction) (16). Furthermore, although radiobiologic models exist to equate doses between various isotopes used for low-dose-rate brachytherapy (17), models that render equivalent doses to high-dose-per-fraction SRT are not available. This point is highlighted, for example, by considering that the mean dose to the retina 5 mm from the tumor edge in the SRT group was 60.1 Gy in 5 fractions, which is equivalent to a biologically equivalent dose of 421.3 Gy<sub>2</sub> (based on an  $\alpha/\beta = 2$ ) and can also be expressed as an equivalent dose in 2 Gy per day fractions of 210.7 Gy<sub>2/2</sub>. These doses would not be compatible with normal tissue survival, and this reflects the challenge we face in modeling these extreme doses. Nevertheless our dosimetric analysis can be interpreted with the consideration that the total dose calculated at each point of interest, which is a surrogate of its respective late toxicity, is specific to each technique. Additionally, dose delivery is under different conditions such that the intra-fraction motion, despite monitoring, is greater with the SRT technique. Each technique demonstrates a specific dose gradient profile, such that IBT yields a steeper dose decline so that the surrounding tissues receive a relatively much lower dose than that prescribed, in comparison with SRT ([Fig. 3](#)).

Given this, our study demonstrated that both IBT and SRT result in high rates of radiation retinopathy and papillophth. This is in contrast to the reported rates after radiation therapy for melanoma in other locations within the ocular fundus (18-21). With respect to the ocular anterior segment structures, both cohorts showed high rates of radiation cataract. This was despite relatively low absolute doses to the lens and the low-dose-rate exposure inherent to IBT. We postulate that this is due to the low threshold for cataract formation inherent to radiation and the lens (22, 23).

Remarkably, we observed significantly greater rates of NVG in the SRT cohort as compared with IBT, despite the small radiation dose reaching anterior segment structures in both techniques. We postulate that this is likely a reflection of greater ischemic changes occurring within the posterior segment tissue rather than

direct radiation insult to the anterior segment vasculature. This is also linked to the observation of greater rates of radiation-induced retinopathy and papillophth observed within the SRT versus IBT cohort. A previous report correlated the histopathology of NVG to the dosimetry in patients treated with SRT (24). That report revealed that radiation-induced damage to the retina is likely causative to radiation-induced NVG. It was hypothesized that the radiation-induced ischemic retinopathy produces angiogenic factors, such as vascular endothelial growth factor, which diffuses through the vitreous to induce vascular proliferative changes in the iris and angle and subsequent development of NVG. This has challenged the dogma that the direct radiation-induced damage to the anterior segment structures is causative. In conclusion, the treatment of juxtapapillary choroidal melanoma with SRT and IBT is associated with high rates of toxicity; SRT, however, produced statistically higher rates of late-tissue toxicity effects, particularly NVG with its devastating visual consequences. Longer follow-up and larger numbers of patients could further validate the results of this study. On the basis of the comparison of the dose distributions at predefined intraocular points of interest, the greater rates of radiation-induced toxicity in the SRT cohort are likely explained by the exposure of the optic nerve and adjacent retina to high radiation dose akin to the prescribed treatment dose, delivered in 5 fractions, in comparison with a relatively smaller dose of the prescribed IBT dose reaching the



**Fig. 3.** Comparison of the dose gradient profile between  $^{125}\text{I}$  brachytherapy (IBT) and stereotactic radiation therapy (SRT), showing the mean dose at points of interest, illustrating steeper fall-off from the prescribed dose with IBT, and intraocular structures receiving significantly lower dose, unlike with SRT.

same structures, delivered at a low exposure rate. The higher incidence of radiation-induced NVG is likely secondary to the more extensive radiation-induced ischemic changes in the ocular posterior segment in SRT. Considering the outcomes in related studies (5, 25), reduction of SRT total dose and dose per fraction, or using single low-dose Gamma Knife radiosurgery, could reduce the long-term radiation toxicities consequent to our current SRT approach.

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